Martingales

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Vedant's Part

What are "fair games"? Mathematically speaking they are games where, on average, your expected wealth tomorrow, given everything you know today, is just your wealth today. Martingales are a way to model these "fair games". In case of super and submartingales they are unfavourable/favourable games respectively.

Definition of Martingale

Let $\mathcal{F}_n$ be a filtration, i.e., an increasing sequence of σ-fields. A sequence X_n is said to be **adapted** to $\mathcal{F}_n$ if $X_n \in \mathcal{F}_n$ for all n.

If X_n is a sequence satisfying:

1. $E|X_n| < \infty$,
2. X_n is adapted to $\mathcal{F}_n$,
3. $E(X_{n+1} \mid \mathcal{F}_n) = X_n$ for all n,

then X is said to be a **martingale** (with respect to $\mathcal{F}_n$).

If in the last definition, $=$ is replaced by $\leq$ or $\geq$, then X is said to be a **supermartingale** or **submartingale**, respectively.

1 Simple Properties of Martingales

Before moving onto more advanced theorems let's first look at a few simple properties of Martingales.

Theorem 1.1. *If X_n is a supermartingale then for $n > m$, $E(X_n \mid \mathcal{F}_m) \leq X_m$.*

Proof. The definition gives the result for $n = m+1$. Suppose $n = m+k$ with $k \geq 2$. By standard properties of conditional expectation (the tower property),

$$E(X_{m+k} \mid \mathcal{F}_m) = E(E(X_{m+k} \mid \mathcal{F}_{m+k-1}) \mid \mathcal{F}_m) \leq E(X_{m+k-1} \mid \mathcal{F}_m)$$

by the definition and inductive hypothesis. The desired result now follows by induction. □

Theorem 1.2.

(i) If X_n is a submartingale then for $n > m$, $E(X_n \mid \mathcal{F}_m) \geq X_m$.

(ii) If X_n is a martingale then for $n > m$, $E(X_n \mid \mathcal{F}_m) = X_m$.

Proof. To prove (i), note that $-X_n$ is a supermartingale and use the previous theorem. For (ii), observe that X_n is a supermartingale and a submartingale. □

This shows us that the definition of a martingale is equivalent if the $n+1$ is replaced by any $s \geq n$.

Theorem 1.3. *If X_n is a martingale w.r.t. $\mathcal{F}_n$ and φ is a convex function with $E|\varphi(X_n)| < \infty$ for all n then $\varphi(X_n)$ is a submartingale w.r.t. $\mathcal{F}_n$. Consequently, if $p \geq 1$ and $E|X_n|^p < \infty$ for all n, then $|X_n|^p$ is a submartingale w.r.t. $\mathcal{F}_n$.*

Proof. By Jensen's inequality and the definition,

$$E(\varphi(X_{n+1}) \mid \mathcal{F}_n) \geq \varphi(E(X_{n+1} \mid \mathcal{F}_n)) = \varphi(X_n).$$

□

Theorem 1.4. *If X_n is a submartingale w.r.t. $\mathcal{F}_n$ and φ is an increasing convex function with $E|\varphi(X_n)| < \infty$ for all n, then $\varphi(X_n)$ is a submartingale w.r.t. $\mathcal{F}_n$. Consequently (i) If X_n is a submartingale then $(X_n - a)^+$ is a submartingale. (ii) If X_n is a supermartingale then $X_n \wedge a$ is a supermartingale.*

Proof. By Jensen's inequality and the assumptions,

$$E(\varphi(X_{n+1}) \mid \mathcal{F}_n) \geq \varphi(E(X_{n+1} \mid \mathcal{F}_n)) \geq \varphi(X_n).$$

□

Let $\mathcal{F}_n, n \geq 0$ be a filtration. $H_n, n \geq 1$ is said to be a **predictable sequence** if $H_n \in \mathcal{F}_{n-1}$ for all $n \geq 1$. In words, the value of H_n may be predicted (with certainty) from the information available at time $n-1$. In this section, we will be thinking of H_n as the amount of money a gambler will bet at time n. This can be based on the outcomes at times $1, \ldots, n-1$ but not on the outcome at time n!

Once we start thinking of H_n as a gambling system, it is natural to ask how much money we would make if we used it. Let X_n be the net amount of money you would have won at time n if you had bet one dollar each time. If you bet according to a gambling system H then your winnings at time n would be

$$(H \cdot X)_n = \sum_{m=1}^{n} H_m(X_m - X_{m-1})$$

since if at time m you have wagered \$3 the change in your fortune would be 3 time that of a person who wagered \$1. Alternatively you can think of X_m is the value of a stock and H_m the number of shares you hold from time $m-1$ to time m.

Suppose now that $\xi_m = X_m - X_{m-1}$ have $P(\xi_m = 1) = p$ and $P(\xi_m = -1) = 1 - p$. A famous gambling system called the "martingale" is defined by $H_1 = 1$ and for $n \geq 2$,

$$H_n = \begin{cases} 2H_{n-1} & \text{if } \xi_{n-1} = -1 \\ 1 & \text{if } \xi_{n-1} = 1 \end{cases}$$

In words, we double our bet when we lose, so that if we lose k times and then win, our net winnings will be 1. To see this consider the following concrete situation

H_n	1	2	4	8	16
ξ_n	-1	-1	-1	-1	1
$(H \cdot X)_n$	-1	-3	-7	-15	1

This system seems to provide us with a "sure thing" as long as $P(\xi_m = 1) > 0$. But in real life expecting this to make you a fortune is quite unrealistic as the amount you have to gamble rises rapidly and most casinos have a table limit! The next result says there is no system for beating an unfavorable game.

Theorem 1.5. *Let $X_n, n \geq 0$, be a supermartingale. If $H_n \geq 0$ is predictable and each H_n is bounded then $(H \cdot X)_n$ is a supermartingale.*

Proof. Using the fact that conditional expectation is linear, $(H \cdot X)_n \in \mathcal{F}_n$, $H_n \in \mathcal{F}_{n-1}$, and properties of conditional expectation, we have

$$\begin{aligned} E((H \cdot X)_{n+1} \mid \mathcal{F}_n) &= (H \cdot X)_n + E(H_{n+1}(X_{n+1} - X_n) \mid \mathcal{F}_n) \\ &= (H \cdot X)_n + H_{n+1}E((X_{n+1} - X_n) \mid \mathcal{F}_n) \leq (H \cdot X)_n \end{aligned}$$

since $E((X_{n+1} - X_n) \mid \mathcal{F}_n) \leq 0$ and $H_{n+1} \geq 0$. □

Remark 1.1. The same result is obviously true for submartingales and for martingales (in the last case, without the restriction $H_n \geq 0$).

We will now consider a very special gambling system: bet \$1 at each time $n \leq N$ then stop playing. A random variable N is said to be a **stopping time** if $\{N = n\} \in \mathcal{F}_n$ for all $n < \infty$, i.e., the decision to stop at time n must be measurable with respect to the information known at that time. If we let $H_n = 1_{\{N \geq n\}}$, then $\{N \geq n\} = \{N \leq n-1\}^c \in \mathcal{F}_{n-1}$, so H_n is predictable, and it follows from the previous theorem that $(H \cdot X)_n = X_{N \wedge n} - X_0$ is a supermartingale. Since the constant sequence $Y_n = X_0$ is a supermartingale and the sum of two supermartingales is also, we have:

Theorem 1.6. *If N is a stopping time and X_n is a supermartingale, then $X_{N \wedge n}$ is a supermartingale.*

Here, U_n and the Upcrossing inequality are defined as in class.

Theorem 1.7 (Martingale Convergence Theorem)**.** *If X_n is a submartingale with* $\sup EX_n^+ < \infty$ *then as $n \to \infty$, X_n converges a.s. to a limit X with $E|X| < \infty$.*

Proof. Since $(X-a)^+ \leq X^+ + |a|$, the Upcrossing Inequality implies that

$$EU_n \leq (|a| + EX_n^+)/(b-a)$$

As $n \uparrow \infty$, $U_n \uparrow U$ the number of upcrossings of $[a, b]$ by the whole sequence, so if $\sup EX_n^+ < \infty$ then $EU < \infty$ and hence $U < \infty$ a.s. Since the last conclusion holds for all rational a and b,

$$\cup_{a,b \in \mathbb{Q}} \{\liminf X_n < a < b < \limsup X_n\} \quad \text{has probability } 0$$

and hence $\limsup X_n = \liminf X_n$ a.s., i.e., $\lim X_n$ exists a.s. Fatou's lemma guarantees $EX^+ \leq \liminf EX_n^+ < \infty$, so $X < \infty$ a.s. To see $X > -\infty$, we observe that

$$EX_n^- = EX_n^+ - EX_n \leq EX_n^+ - EX_0$$

(since X_n is a submartingale), so another application of Fatou's lemma shows

$$EX^- \leq \liminf_{n \to \infty} EX_n^- \leq \sup_n EX_n^+ - EX_0 < \infty$$

and completes the proof. □

Theorem 1.8. *If $X_n \geq 0$ is a supermartingale then as $n \to \infty$, $X_n \to X$ a.s. and $EX \leq EX_0$.*

Proof. $Y_n = -X_n \leq 0$ is a submartingale with $EY_n^+ = 0$. Since $EX_0 \geq EX_n$, the inequality follows from Fatou's lemma. □

1.1 Examples

Example 1.1. Let μ be a finite measure and ν be a probability measure on $(\Omega, \mathcal{F})$. Let $\mathcal{F}_n \uparrow \mathcal{F}$ be σ-fields ($\sigma(\cup \mathcal{F}_n) = \mathcal{F}$). Let μ_n and ν_n be restrictions of μ and ν to $\mathcal{F}_n$.

Suppose $\mu_n \ll \nu_n$ for all n.

$$X_n = \frac{d\mu_n}{d\nu_n} \quad \text{then } X_n \text{ is a martingale w.r.t } \mathcal{F}_n.$$

Let $A \in \mathcal{F}_n$. Since $X_n \in \mathcal{F}_n$ and ν_n is restriction of ν to $\mathcal{F}_n$,

$$\int_A X_n d\nu = \int_A X_n d\nu_n = \mu_n(A) = \mu(A)$$

Now consider $Z = E[Y \mid \mathcal{G}]$, for $A \in \mathcal{G}$, $\int_A Z d\nu = \int_A Y d\nu$. Let $A \in \mathcal{F}_{n-1} \subset \mathcal{F}_n$.

$$\int_A X_n d\nu = \mu(A) = \int_A X_{n-1} d\nu$$

$$\therefore E(X_n \mid \mathcal{F}_{n-1}) = X_{n-1}$$

Example 1.2 (Branching Process (Galton-Watson))**.** Let $\xi_i^n, i, n \geq 1$ be i.i.d. non-negative random variables. Let $Z_0 = 1$ and

$$Z_{n+1} = \begin{cases} \xi_1^{n+1} + \cdots + \xi_{Z_n}^{n+1} & Z_n > 0 \\ 0 & Z_n = 0 \end{cases}$$

Then if $\mathcal{F}_n = \sigma(\xi_i^m : i \geq 1, 1 \leq m \leq n)$ and $\mu = E\xi_i^m$, then Z_n/μ^n is a $\mathcal{F}_n$ martingale.

$$E(Z_{n+1} \mid \mathcal{F}_n) = E(\xi_1^{n+1} + \cdots + \xi_{Z_n}^{n+1}) = Z_n\mu = \mu Z_n$$

$$\therefore \frac{Z_n}{\mu^n} \text{ is a martingale.}$$

$p_k = P(\xi_i^m = k)$ is the offspring distribution. The limit exists but usually not trivial but let's look at cases when it is.

If $\mu < 1$ and then $Z_n = 0$ for all $n \gg$ large, so $Z_n/\mu^n \to 0$.

$$E(Z_n/\mu^n) = E(Z_0) = 1 \implies E(Z_n) = \mu^n$$

$$P(Z_n > 0) \leq E(Z_n) = \mu^n \to 0 \quad \text{if } \mu < 1.$$

Intuitively, species dies out.

Example 1.3. This example shows why the hypothesis for Martingale Convergence theorem does not guarantee convergence in L^1. Let S_n be a symmetric simple random walk with $S_0 = 1$, i.e., $S_n = S_{n-1} + \xi_n$ where $\xi_1, \xi_2, \ldots$ are i.i.d. with $P(\xi_i = 1) = P(\xi_i = -1) = 1/2$. Let $N = \inf\{n : S_n = 0\}$ and let $X_n = S_{N \wedge n}$. Earlier results imply that X_n is a nonnegative martingale. The Martingale Convergence Theorem implies X_n converges to a limit $X_\infty < \infty$ that must be $\equiv 0$, since convergence to $k > 0$ is impossible. (If $X_n = k > 0$ then $X_{n+1} = k \pm 1$.) Since $EX_n = EX_0 = 1$ for all n and $X_\infty = 0$, convergence cannot occur in L^1.

2 L^p Convergence Theorem and Bounds on Square Integrable Martingales

Theorem 2.1 (L^p Convergence Theorem)**.** *If X_n is a martingale with $\sup E|X_n|^p < \infty$, $p > 1$, then $X_n \to X$ a.s. and in L^p.*

Proof. By Jensen's inequality,

$$(EX_n^+)^p \leq (E|X_n|)^p \leq E|X_n|^p < C$$

so

$$\sup EX_n^+ \leq C^{1/p}$$

Thus by the Martingale Convergence Theorem, $X_n \to X$ a.s. Also by Doob's Maximal Inequality,

$$\begin{aligned} E\left[\left(\sup_{0 \leq m \leq n} |X_m|\right)^p\right] &\leq \left(\frac{p}{p-1}\right)^p E|X_n|^p \\ &\leq \left(\frac{p}{p-1}\right)^p C \end{aligned}$$

Define $Y_n = \left(\sup_{0 \leq m \leq n} |X_m|\right)^p$, which is non-decreasing. Let $Y = \lim_{n\to\infty} Y_n = \left(\sup_{m \geq 0} |X_m|\right)^p$. By MCT,

$$\lim_{n\to\infty} E[Y_n] = E[\lim_{n\to\infty} Y_n] = E[Y]$$

This means
$$\lim_{n\to\infty} E\left[\left(\sup_{0\le m\le n}|X_m|\right)^p\right] = E\left[\left(\sup_{m\ge 0}|X_m|\right)^p\right] \le \left(\frac{p}{p-1}\right)^p C$$
Therefore, $S = \sup_{m\ge 0}|X_m| \in L^p$.

Let $Z_n = |X_n - X|^p$. As $X_n \to X$ a.s., $X_n - X \to 0$ a.s., thus $Z_n \to 0$ a.s. Also
$$|X_n| \le S \quad \forall n$$
$$|X| \le S$$
Therefore, $|X - X_n| \le |X_n| + |X| \le S + S = 2S$.
$$|X - X_n|^p \le 2^p S^p$$
Let $G = 2^p S^p$.
$$E[G] = 2^p E[S^p] < \infty \implies G \in L^1$$
By DCT, since $Z_n \to 0$ a.s. and $|Z_n| \le G$ with $E[G] < \infty$,
$$\lim_{n\to\infty} E[Z_n] = E[\lim_{n\to\infty} Z_n] = E[0] = 0$$
This proves $X_n \to X$ in L^p.

Remark: Why not L^1? Too weak, also $(\frac{p}{1-p})^p \to \infty$ as $p \to 1$. □

X_n is a martingale with $X_0 = 0$ and $EX_n^2 < \infty$ for all n.

X_n^2 is a submartingale. It follows from Doob's decomposition theorem that we can write
$$X_n^2 = M_n + A_n,$$
where M_n is a martingale, and from the standard formulas for Doob decompositions that
$$A_n = \sum_{m=1}^{n}\left(E(X_m^2 \mid \mathcal{F}_{m-1}) - X_{m-1}^2\right) = \sum_{m=1}^{n} E\left((X_m - X_{m-1})^2 \mid \mathcal{F}_{m-1}\right).$$
In particular,
$$A_\infty = \sum_{m=1}^{\infty} E\left((X_m - X_{m-1})^2 \mid \mathcal{F}_{m-1}\right)$$
is the *total conditional variance.*

Theorem 2.2.
$$E\left(\sup_m |X_m|^2\right) \le 4EA_\infty$$
Proof. Using Doob's inequality with $p = 2$,
$$E\left[\left(\sup_{0\le m\le n}|X_m|^2\right)\right] \le 4E[E|X_n|^2]$$
$$EX_n^2 = EA_n + EM_n = EA_n + EM_0$$
Since $M_0 = X_0^2 - A_0$, therefore $EM_0 = 0$. Thus by MCT, taking limits, the result follows. □

Theorem 2.3. *Let X_n be a square integrable martingale with $X_0 = 0$. Then $\lim_{n\to\infty} X_n$ exists and is finite almost surely on the set $\{A_\infty < \infty\}$.*

Proof. Define the stopping time $\tau = \inf\{n : A_{n+1} > a^2\}$. Consider the stopped martingale $Y_n = X_{\tau\wedge n}$. From the previous theorem, we have the bound:

$$E\left(\sup_n |X_{\tau\wedge n}|^2\right) \le 4E[A_\tau] \le 4a^2.$$

Since the supremum is L^2 bounded (and thus L^1 bounded), by the Martingale Convergence Theorem (or L^2 convergence theorem logic), $\lim_{n\to\infty} X_{\tau\wedge n}$ exists and is finite a.s.

Let $\Omega_a = \{A_\infty \le a^2\}$. On this set, $A_{n+1} \le a^2$ for all n, which implies $\tau = \infty$. Thus, on Ω_a, $X_{\tau\wedge n} = X_n$ for all n. Consequently, X_n converges on Ω_a. Since $\{A_\infty < \infty\} = \bigcup_{k=1}^\infty \Omega_k$ (taking $a = k$), the result follows. □

Theorem 2.4.

$$E\left(\sup_n |X_n|\right) \le 3EA_\infty^{1/2}$$

Proof. We know that for a non-negative random variable Y,

$$E[Y] = \int_0^\infty P(Y > a)da$$

We have $\sup_n |X_n|$. We want to bound $P(\sup_n |X_n| > a)$.

Define $\tau = \inf\{n \mid A_{n+1} > a^2\}$.

$$\{A_\infty \text{ exceeds } a^2\} = \{A_\infty > a^2\}$$

$$P(A \cup B) \le P(A) + P(B)$$

$$\begin{aligned} P(\sup_n |X_n| > a) &\le P(A_\infty > a^2) + P(\sup_n |X_n| > a \text{ and } \tau = \infty) \\ &\le P(A_\infty > a^2) + P(\sup_m |X_{\tau\wedge m}| > a) \end{aligned}$$

since if $\tau = \infty$, then $X_{\tau\wedge m} = X_m$.

For the second term,

$$P(\sup_m |X_{\tau\wedge m}| > a) \le \frac{E[X_{\tau\wedge n}^2]}{a^2}$$

$$E[X_{\tau\wedge n}^2] = E[A_{\tau\wedge n}], \quad A_{\tau\wedge n} \le A_\infty \wedge a^2$$

So,

$$P(\sup_m |X_{\tau\wedge m}| > a) \le \frac{E[A_\infty \wedge a^2]}{a^2}$$

Therefore,

$$\begin{aligned} E[\sup_m |X_m|] &= \int_0^\infty P(\sup_m |X_m| > a)da \\ &\le \int_0^\infty P(A_\infty > a^2)da + \int_0^\infty \frac{1}{a^2} E[A_\infty \wedge a^2]da \end{aligned}$$

Let $Y = A_\infty^{1/2}$, then $\{A_\infty > a^2\} = \{Y > a\}$.

$$\int_0^\infty P(Y > a)da = E[Y] = E[A_\infty^{1/2}]$$

For the second term, using Fubini's theorem,

$$\int_0^\infty \frac{1}{a^2} E[A_\infty \wedge a^2]da = E\left[\int_0^\infty \frac{1}{a^2}(A_\infty \wedge a^2)da\right]$$

Let $C = A_\infty$. Split integral at $\sqrt{C} = a$. 1. For $a \in [0, \sqrt{C}]$, $a^2 \leq C$ so $C \wedge a^2 = a^2$.

$$\int_0^{\sqrt{C}} a^{-2}(a^2)da = \int_0^{\sqrt{C}} 1da = \sqrt{C}$$

2. For $a \in [\sqrt{C}, \infty]$, $a^2 > C$ so $C \wedge a^2 = C$.

$$C \int_{\sqrt{C}}^{\infty} \frac{1}{a^2} da = C \left[-a^{-1}\right]_{\sqrt{C}}^{\infty} = C \frac{1}{\sqrt{C}} = \sqrt{C}$$

Thus,

$$E[\sqrt{C} + \sqrt{C}] = 2E[A_\infty^{1/2}]$$

Total bound is $E[A_\infty^{1/2}] + 2E[A_\infty^{1/2}] = 3E[A_\infty^{1/2}]$. Hence proved. $\square$

The above theorem shows us that we can bound the L^1 norm of a maximum by an $L^{1/2}$ norm.

3 Local Martingales

3.1 Motivation

While martingales form the backbone of stochastic analysis, in continuous time it is often too restrictive to require that the martingale property holds globally for all time without additional integrability conditions. Many natural stochastic processes, including Itô integrals such as

$$M_t = \int_0^t \varphi_s dW_s$$

for merely locally square-integrable predictable φ, need not be integrable on the entire horizon, but are still "locally" martingales: they behave like martingales up to random times that prevent explosion or loss of integrability.

This motivates the concept of local martingales, which generalize martingales while preserving most of their key properties.

3.2 Definition

Definition 3.1 (Local martingale)**.** Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\geq 0}, \mathbb{P})$ be a filtered probability space satisfying the usual hypotheses. An adapted càdlàg process $M = (M_t)_{t\geq 0}$ is called a local martingale if there exists a sequence of stopping times $(\tau_n)_{n\geq 1}$ with

$$\tau_n \uparrow \infty \quad \text{a.s.}$$

such that for each n, the stopped process $M^{\tau_n} = (M_{t\wedge\tau_n})_{t\geq 0}$ is a (true) martingale.

Remark 3.1. The sequence (τ_n) is called a *localizing sequence*. A process M is a martingale if and only if it is a local martingale with a constant localizing sequence $\tau_n \equiv \infty$.

Example 3.1.

1. Any (true) martingale is trivially a local martingale.
2. The process $M_t = \int_0^t \varphi_s dW_s$ is a local martingale for any predictable φ satisfying $\int_0^t \varphi_s^2 ds < \infty$ for all t.
3. The stochastic exponential $Z_t = \exp(W_t - \frac{1}{2}t)$ is a nonnegative martingale (hence a local martingale). In contrast, processes such as $Z_t = \exp(\int_0^t \theta_s dW_s - \frac{1}{2}\int_0^t \theta_s^2 ds)$ may be only local martingales if integrability fails.

3.3 Localization principle

Proposition 3.1 (Localization preserves martingale property)**.** *If M is a martingale and τ is a stopping time, then the stopped process $M^\tau = (M_{t\wedge\tau})_{t\geq 0}$ is a martingale.*

Proof. Fix $s < t$. Since τ is a stopping time, the event $\{\tau \leq s\}$ is $\mathcal{F}_s$-measurable. Then

$$\mathbb{E}[M_{t\wedge\tau}|\mathcal{F}_s] = \mathbb{E}[M_t I_{\{\tau>t\}} + M_\tau I_{\{\tau\leq t\}}|\mathcal{F}_s].$$

On $\{\tau \leq s\}$, $M_{t\wedge\tau} = M_\tau = M_{s\wedge\tau}$. On $\{\tau > s\}$, $\mathbb{E}[M_{t\wedge\tau}|\mathcal{F}_s] = \mathbb{E}[M_t|\mathcal{F}_s] = M_s = M_{s\wedge\tau}$. Combine both to get $\mathbb{E}[M_{t\wedge\tau}|\mathcal{F}_s] = M_{s\wedge\tau}$. □

Proposition 3.2 (Stopping and localization)**.** *If M is a local martingale and τ a stopping time, then the stopped process M^τ is also a local martingale.*

Proof. If (τ_n) is a localizing sequence for M, then $(\tau_n \wedge \tau)$ is a localizing sequence for M^τ, because each stopped version $(M^\tau)^{\tau_n\wedge\tau} = M^{\tau_n\wedge\tau}$ is a true martingale. □

3.4 Integrability and the true martingale property

A local martingale need not be a martingale globally, since integrability might fail beyond the localization sequence. The following gives a key criterion.

Theorem 3.1 (Local martingale dominated in L^1 is a martingale)**.** *Let M be a local martingale on $[0,T]$. If*

$$\sup_{0\le t\le T} \mathbb{E}[|M_t|] < \infty,$$

then M is a true martingale on $[0,T]$.

Proof. Let (τ_n) be a localizing sequence with M^{τ_n} martingales. For $0 \le s < t \le T$, $\mathbb{E}[M_{t\wedge\tau_n} \mid \mathcal{F}_s] = M_{s\wedge\tau_n}$. Because $|M_{t\wedge\tau_n}| \le \sup_{u\le T} |M_u|$ and the latter has finite expectation, we may apply dominated convergence to conditional expectations:

$$\mathbb{E}[M_t|\mathcal{F}_s] = \lim_{n\to\infty} \mathbb{E}[M_{t\wedge\tau_n}|\mathcal{F}_s] = \lim_{n\to\infty} M_{s\wedge\tau_n} = M_s.$$

Hence M is a martingale. □

3.5 Nonnegative local martingales

Proposition 3.3 (Nonnegative local martingales are supermartingales)**.** *If $M \ge 0$ is a local martingale, then $\mathbb{E}[M_t]$ is nonincreasing in t, and M is a supermartingale.*

Proof. Let (τ_n) localize M. Each M^{τ_n} is a nonnegative martingale, so $\mathbb{E}[M_{t\wedge\tau_n}]$ is constant in t. As $n \to \infty$, by monotone convergence $\mathbb{E}[M_t] \le \mathbb{E}[M_s]$ for $s < t$. Therefore $\mathbb{E}[M_t|\mathcal{F}_s] \le M_s$, i.e., M is a supermartingale. □

Corollary 3.1 (Limit exists)**.** *If $M \ge 0$ is a local martingale, then the limit*

$$M_\infty := \lim_{t\to\infty} M_t$$

exists a.s. (finite or zero).

Proof. Nonnegative supermartingales converge a.s. by the monotone convergence theorem for conditional expectations or Doob's martingale convergence theorem. □

Remark 3.2. A nonnegative local martingale need not be a martingale; it can be a strict local martingale if $\mathbb{E}[M_t] < \mathbb{E}[M_0]$ for $t > 0$. Examples arise from inverse Bessel processes or certain exponential Brownian functionals in mathematical finance.

3.6 Examples of localization

Example 3.2. Let $M_t = \int_0^t \frac{1}{1+|W_s|} dW_s$. Although the integrand is not square-integrable globally, it is on each finite interval. For each n, set $\tau_n = \inf\{t : |W_t| \ge n\}$. Then M^{τ_n} is a bounded Itô integral and hence a martingale. Therefore M is a local martingale.

Example 3.3. Let $Z_t = \exp(\theta W_t - \frac{1}{2}\theta^2 t)$ for constant θ. Then Z is a true martingale (by the exponential martingale theorem). If we replace θ by an unbounded predictable process (say $\theta_t = W_t$), the resulting process is only a local martingale, since exponential integrability fails.

3.7 Strict local martingales

Definition 3.2. A local martingale M is called a *strict local martingale* if it is not a true martingale, i.e., $\mathbb{E}[M_t] < \mathbb{E}[M_0]$ for some $t > 0$.

Example 3.4 (A strict local martingale)**.** Let $X_t = 1/R_t$ where R_t is a 3-dimensional Bessel process (distance of 3D Brownian motion from the origin). Then X_t is a positive local martingale but not a true martingale: $\mathbb{E}[X_t] < X_0$ for all $t > 0$. This process plays a key role in examples of "financial bubbles".

3.8 Continuous local martingales and quadratic variation

Theorem 3.2 (Quadratic variation)**.** *If M is a continuous local martingale, there exists an increasing adapted process $[M]_t$ (the quadratic variation) such that*

$$M^2 - [M]$$

is a local martingale. Moreover, for any two continuous local martingales M, N, there exists a covariation process $[M, N]$ with

$$MN - [M, N]$$

a local martingale.

Idea. The result follows from the Itô construction and limiting argument with simple predictable approximations of the integrand in the stochastic integral defining M. For Brownian motion, $[W]_t = t$. Linearity and the polarization identity yield the general case.

Remark 3.3. For semimartingales $X = M + A$ the quadratic variation comes entirely from the local martingale part: $[X] = [M]$. This property was used in the definition of semimartingales in the previous section.

3.9 Important properties summarized

1. The class of local martingales is a vector space.
2. Stopping a local martingale gives another local martingale.
3. Every local martingale is a semimartingale.
4. Nonnegative local martingales are supermartingales.
5. Continuous finite-variation local martingales are constant.
6. Quadratic variation exists and behaves as in the Brownian case.

3.10 Exercises (practice and intuition)

1. **(Checking definition)** Show that for Brownian motion W, the process $M_t = \int_0^t \sin(W_s)dW_s$ is a local martingale.
   Hint: The integrand is bounded; thus the integral is square-integrable on each finite interval.
2. **(Localization)** Let $M_t = \int_0^t W_s^3 dW_s$. Construct a localization sequence making M a local martingale.
   Hint: Use $\tau_n = \inf\{t : |W_t| \geq n\}$ and boundedness of $W_{s\wedge\tau_n}$.
3. **(Nonnegative local martingale)** Show that $Z_t = \exp(W_t - \frac{1}{2}t)$ satisfies $\mathbb{E}[Z_t] = 1$. Then consider $Z_t = \exp(W_t^2 - t)$; verify it is a local martingale but not a martingale.
4. **(Quadratic variation)** Compute $[M]_t$ for $M_t = \int_0^t e^s dW_s$.
   Solution: $[M]_t = \int_0^t e^{2s}ds = \frac{1}{2}(e^{2t} - 1)$.
5. **(Strict local martingale)** Research and describe one explicit example of a strict local martingale and explain why it fails to be a true martingale.

3.11 Summary Table

Type	**Characterization**
Martingale	$\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s$, integrable for all t
Local martingale	$\exists \tau_n \uparrow \infty$ s.t. M^{τ_n} are martingales
Strict local martingale	Local martingale but not martingale ($\mathbb{E}[M_t] < M_0$)
Nonnegative local martingale	Supermartingale, $\mathbb{E}[M_t] \le M_0$
Continuous local martingale	Has quadratic variation $[M]$, $M_t^2 - [M]_t$ local martingale

Theorem 3.3 (Dubins–Schwarz Theorem). *Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\ge 0}, P)$ be a filtered probability space satisfying the usual conditions. Let M_t be a continuous local martingale with $M_0 = 0$, and let $[M]_t$ denote its quadratic variation. Assume that*

$$[M]_\infty = \infty \qquad a.s.$$

(i.e. the quadratic variation increases without bound). Define the random time-change

$$\tau_t := \inf\{s \ge 0 : [M]_s > t\}, \qquad t \ge 0.$$

Then:

1. *The process*
$$B_t := M_{\tau_t}, \qquad t \ge 0,$$
adapted to the filtration $\mathcal{G}_t := \mathcal{F}_{\tau_t}$, is a standard Brownian motion.

2. *The original local martingale M_t can be recovered by the relation*
$$M_t = B_{[M]_t}, \qquad t \ge 0.$$

Proof. We outline the standard argument.

Step 1: Time change. Because $[M]_t$ is continuous, strictly increasing, and diverges to $+\infty$, its inverse

$$\tau_t = \inf\{s : [M]_s > t\}$$

is finite and continuous for all $t \ge 0$. Define $B_t := M_{\tau_t}$ and $\mathcal{G}_t := \mathcal{F}_{\tau_t}$.

Step 2: B_t is a continuous local martingale. A standard theorem on time-changes of continuous local martingales states that if M_t is a continuous local martingale and τ_t is a continuous increasing family of stopping times, then M_{τ_t} is a continuous local martingale with respect to the time-changed filtration. Thus B_t is a continuous local martingale with $B_0 = 0$.

Step 3: Quadratic variation of B_t. Quadratic variation respects time changes:

$$[B]_t = [M_\tau]_t = [M]_{\tau_t}.$$

Since τ_t is the inverse of $[M]_s$, we have $[M]_{\tau_t} = t$. Therefore

$$[B]_t = t.$$

Step 4: Apply Lévy's Characterization. By Lévy's theorem, a continuous local martingale starting at 0 whose quadratic variation is t is a standard Brownian motion. Thus B_t is a Brownian motion.

Step 5: Recovering M_t. From the definition of B_t,

$$B_{[M]_t} = M_{\tau_{[M]_t}}.$$

Since τ is the inverse of $[M]$, we have $\tau_{[M]_t} = t$, hence

$$B_{[M]_t} = M_t.$$

□

Interpretation. The Dubins–Schwarz theorem makes the connection between a continuous local martingale M_t and its quadratic variation $[M]_t$ concrete. The identity

$$M_t = B_{[M]_t}$$

shows that quadratic variation is not merely an abstract quantity: it acts as the *intrinsic time* or *operational clock* of the martingale. The process M_t evolves only when its quadratic variation increases; in other words, $[M]_t$ measures exactly how much "martingale activity" has accumulated up to time t. Thus the path of M can be viewed as a standard Brownian motion B run on its own internal clock $[M]_t$.

Concluding remark. Local martingales play the same role in continuous-time stochastic calculus as martingales do in discrete time: they provide the building blocks for semimartingales and Itô calculus. All integrals, SDEs, and representation results are ultimately expressed in terms of local martingales.

Aleena's Part

4 Semimartingales

4.1 Motivation

The class of semimartingales is the most general and robust family of stochastic processes for which the stochastic integral can be defined consistently. Every Itô integral we have constructed so far involved a Brownian motion W or, more generally, a continuous local martingale M. However, in many applications (e.g. stochastic differential equations with drift, stochastic control, finance) we encounter processes that are not martingales themselves but rather the sum of a martingale and a finite-variation part.

Semimartingales arise naturally because real-world processes typically combine both unpredictable fluctuations and systematic trends. In finance, asset prices exhibit random shocks alongside deterministic growth; in physics, deterministic forces interact with thermal noise. The semimartingale decomposition $X = M + A$ captures this structure precisely; M represents the "noise part" with martingale-like behavior, while A accounts for accumulated drift or other regular effects. This decomposition is what makes semimartingales both flexible enough to model realistic phenomena and sufficiently regular to support a stable theory of stochastic integration.

Example 4.1. Consider a diffusion

$$X_t = x_0 + \int_0^t b(s, X_s)ds + \int_0^t \sigma(s, X_s)dW_s.$$

The first integral is of finite variation (an ordinary Riemann integral); the second is a local martingale (Itô integral). Thus X is the sum of a finite-variation process and a local martingale: a typical semimartingale.

This decomposition turns out to be exactly what is needed to extend Itô calculus beyond Brownian motion to a large class of processes.

4.2 Definition and first properties

Definition 4.1 (Semimartingale)**.** An adapted càdlàg process $X = (X_t)_{t\geq 0}$ is called a semimartingale if it can be written as

$$X_t = X_0 + M_t + A_t,$$

where M is a local martingale, A is an adapted càdlàg finite-variation process with $A_0 = 0$. The pair (M, A) is called a semimartingale decomposition of X.

Example 4.2.

1. Every finite-variation process $A_t = \int_0^t f(s)ds$ (with f adapted and integrable) is a semimartingale: $M \equiv 0$, A_t itself finite-variation.

2. Every local martingale M is a semimartingale: $A \equiv 0$.

3. Every diffusion $X_t = x_0 + \int_0^t b(s)ds + \int_0^t \sigma(s)dW_s$ is a semimartingale.

4. Any continuous adapted process of bounded quadratic variation admits a unique decomposition $M + A$ and is a continuous semimartingale.

4.3 Finite-variation processes and quadratic variation

Definition 4.2 (Finite-variation process)**.** A càdlàg process A_t is of finite variation on each bounded interval if, for every $T > 0$,

$$V_T(A) := \sup_{\pi} \sum_{i=0}^{n-1} |A_{t_{i+1}} - A_{t_i}| < \infty \quad \text{a.s.},$$

where the supremum runs over all partitions $\pi : 0 = t_0 < t_1 < \cdots < t_n = T$.

Finite-variation processes have zero quadratic variation:

$$[A]_t = \lim_{|\pi| \to 0} \sum |A_{t_{i+1}} - A_{t_i}|^2 = 0.$$

Hence in a semimartingale decomposition $X = M + A$ the quadratic variation of X comes entirely from M:

$$[X]_t = [M]_t.$$

4.4 Quadratic covariation and continuous semimartingales

Definition 4.3 (Quadratic covariation)**.** If X, Y are semimartingales with decompositions $X = M^X + A^X$, $Y = M^Y + A^Y$, the quadratic covariation is defined as

$$[X, Y]_t := [M^X, M^Y]_t,$$

the quadratic covariation of their local martingale parts. For a single process, $[X] := [X, X]$.

Remark 4.1. Finite-variation components do not contribute to quadratic covariation: if A is finite variation and M a local martingale then $[A, M] = 0$. Thus for a semimartingale $X = M + A$, $[X] = [M]$.

4.5 Examples

1. **Deterministic drifted Brownian motion.** Let $X_t = W_t + \mu t$. Then $M_t = W_t$ (local martingale), $A_t = \mu t$ (finite variation). Hence X is a semimartingale.

   W_t is a standard Brownian motion and hence it is a local martingale.

   The process $A_t = \mu t$ has zero quadratic variation on $[0, T]$; in particular it is a finite variation process.

   Fix a partition $\pi = \{0 = t_0 < t_1 < \cdots < t_n = T\}$ of $[0, T]$. The quadratic variation along π is

   $$Q(\pi; A) \;=\; \sum_{i=0}^{n-1} \big(A_{t_{i+1}} - A_{t_i}\big)^2 \;=\; \sum_{i=0}^{n-1} \big(\mu(t_{i+1} - t_i)\big)^2 \;=\; \mu^2 \sum_{i=0}^{n-1} (t_{i+1} - t_i)^2.$$

   Let $|\pi| := \max_i (t_{i+1} - t_i)$ denote the mesh size of the partition. Then

   $$Q(\pi; A) \;=\; \mu^2 \sum_{i=0}^{n-1} (t_{i+1} - t_i)^2 \le \mu^2 \, |\pi| \sum_{i=0}^{n-1} (t_{i+1} - t_i) = \mu^2 \, |\pi| \, T.$$

   Since $|\pi| \to 0$ for a sequence of refining partitions, we obtain

   $$\lim_{|\pi| \to 0} Q(\pi; A) = 0.$$

   Therefore we can conclude X is a semimartingale.

2. **Ornstein-Uhlenbeck process.** The process solving $dX_t = -\theta X_t dt + \sigma dW_t$ has $X_t = X_0 - \theta \int_0^t X_s ds + \sigma W_t$. It is again a semimartingale with local martingale part $M_t = \sigma W_t$ and finite-variation part $A_t = -\theta \int_0^t X_s ds$.

   The stochastic integral is a scaled Brownian motion and therefore a martingale. Since X_t is continuous and bounded on $[0, T]$, we have

   $$\int_0^T |X_s|\, ds < \infty \quad \text{a.s.}$$

   Thus the Ornstein–Uhlenbeck process is a semimartingale.

3. **Poisson process.** For a Poisson process N_t with rate λ, the compensated process $M_t = N_t - \lambda t$ is a martingale. Thus $N_t = M_t + \lambda t$ is a semimartingale with finite-variation drift $A_t = \lambda t$.

   Let N_t be a Poisson process with intensity λ.

   Part 1: $M_t = N_t - \lambda t$ is a martingale.

   $$E[M_t \mid \mathcal{F}_s] = E[N_t - \lambda t \mid \mathcal{F}_s] = E[N_t \mid \mathcal{F}_s] - \lambda t.$$

   Using independent increments:

   $$E[N_t \mid \mathcal{F}_s] = N_s + \lambda(t - s).$$

   Thus:

   $$E[M_t \mid \mathcal{F}_s] = N_s + \lambda(t - s) - \lambda t = N_s - \lambda s = M_s.$$

   Part 2: N_t is a semimartingale because

   $$N_t = M_t + \lambda t,$$

   a local martingale plus finite-variation term.

4.6 Stability properties

Proposition 4.1 (Linearity and stopping)**.**

1. *The set of semimartingales is a vector space: if X, Y are semimartingales, so are $aX + bY$ for constants a, b.*

2. *If X is a semimartingale and τ a stopping time, then the stopped process $X_t^\tau := X_{t\wedge\tau}$ is a semimartingale with decomposition $M^\tau + A^\tau$.*

3. *If H is a bounded predictable process and X a semimartingale, then the stochastic integral $Y_t := \int_0^t H_s dX_s$ is again a semimartingale with decomposition $Y_t = \int_0^t H_s dM_s + \int_0^t H_s dA_s$.*

Proof. **(1) Linearity.** Let X and Y be semimartingales. Then they admit decompositions

$$X = M^X + A^X, \qquad Y = M^Y + A^Y,$$

where M^X, M^Y are local martingales and A^X, A^Y are finite–variation processes. For constants $a, b \in \mathbb{R}$, consider $Z = aX + bY$. Then

$$Z = a(M^X + A^X) + b(M^Y + A^Y) = (aM^X + bM^Y) + (aA^X + bA^Y) =: M^Z + A^Z.$$

Since a linear combination of local martingales is again a local martingale, M^Z is a local martingale. Similarly, A^Z has finite variation because

$$|A^Z_{t_{i+1}} - A^Z_{t_i}| \leq |a| \cdot |A^X_{t_{i+1}} - A^X_{t_i}| + |b| \cdot |A^Y_{t_{i+1}} - A^Y_{t_i}|,$$

and both A^X and A^Y have finite variation. Thus Z is a semimartingale.

(2) Stopping. Let $X = M + A$ be a semimartingale and let τ be a stopping time. Define the stopped process

$$X^\tau_t := X_{t\wedge\tau} = M_{t\wedge\tau} + A_{t\wedge\tau} =: M^\tau_t + A^\tau_t.$$

Since stopping preserves local martingales, M^τ is a local martingale. Finite variation is also preserved, because for any partition π of $[0, t]$,

$$\sum_i |A^\tau_{t_{i+1}} - A^\tau_{t_i}| = \sum_i |A_{t_{i+1}\wedge\tau} - A_{t_i\wedge\tau}| \leq \mathrm{V}_{[0,\mathrm{t}]}(A) < \infty.$$

Thus X^τ is a semimartingale.

(3) Stochastic integration. Let H be a bounded predictable process and let $X = M + A$ be a semimartingale. Define

$$Y_t := \int_0^t H_s \, dX_s.$$

By linearity of the stochastic integral,

$$Y_t = \int_0^t H_s \, dM_s + \int_0^t H_s \, dA_s =: N_t + B_t.$$

Since H is bounded and predictable, $\int H \, dM$ is a local martingale. Moreover, $\int H \, dA$ has finite variation because

$$\sum_i |B_{t_{i+1}} - B_{t_i}| = \sum_i \left| \int_{t_i}^{t_{i+1}} H_s \, dA_s \right| \leq C \ \mathrm{V}_{[0,\mathrm{t}]}(A),$$

where $|H| \leq C$. Thus Y is a semimartingale.

□

4.7 Alternative characterization

Theorem 4.1 (Equivalent characterization)**.** *A càdlàg adapted process X is a semimartingale if and only if for every bounded predictable simple process H, the stochastic integral $\int_0^t H_s dX_s$ can be defined and satisfies the integration-by-parts identity*

$$H_t X_t - H_0 X_0 = \int_0^t H_s dX_s + \int_0^t X_{s-} dH_s + [H, X]_t,$$

where $[H, X]$ is the covariation process.

This equivalence is one of the fundamental theorems in stochastic integration theory. It implies that the class of semimartingales is the largest class of integrators for which Itô calculus (and hence stochastic differential equations) makes sense.

Theorem 4.2 (Integration by parts for semimartingales)**.** *If X, Y are semimartingales then*

$$X_t Y_t = X_0 Y_0 + \int_0^t X_{s-} dY_s + \int_0^t Y_{s-} dX_s + [X, Y]_t.$$

Sketch. Apply the product rule for finite sums on partitions and take the limit; the cross term arises from the quadratic covariation of the martingale parts. The full proof uses the definition of the stochastic integral and the limiting properties of quadratic variation.

4.8 Continuous semimartingales and Itô's formula

A continuous semimartingale $X = M + A$ satisfies M continuous local martingale, A continuous finite variation. For any $f \in C^2(\mathbb{R})$, Itô's formula states

$$f(X_t) = f(X_0) + \int_0^t f'(X_s)dX_s + \frac{1}{2}\int_0^t f''(X_s)d[X]_s$$

where $dX_s = dM_s + dA_s$. This was proved earlier in the Itô calculus chapter and holds for every continuous semimartingale.

Remark 4.2. The term $d[X]_s$ depends only on the martingale part M; finite variation does not contribute to quadratic variation.

4.9 Decomposition and uniqueness

Theorem 4.3 (Uniqueness of semimartingale decomposition)**.** *If $X = M + A = M' + A'$ with M, M' local martingales and A, A' finite-variation adapted processes, then $M = M'$ and $A = A'$ (indistinguishably).*

Proof. Set $N := M - M' = A' - A$. Then N is both a local martingale and a finite-variation process. By the theorem that a continuous finite-variation local martingale is constant, $N_t \equiv N_0$ a.s. Since we can choose decompositions with $M_0 = M'_0 = 0$, we get $N_0 = 0$, hence $N \equiv 0$. Thus $M = M'$ and $A = A'$. □

4.10 Exercises

1. Let $X_t = W_t + t^2$. Identify the martingale and finite-variation parts and compute the quadratic variation.
   Solution:

   We need to decompose $X_t = W_t + t^2$ into the form $X = M + A$ where M is a local martingale and A is a finite-variation process.

   The local martingale part is $M_t = W_t$ (since W_t is a standard brownian motion is a martingale) and the finite-variation part is $A_t = t^2$.

   To show that $A_t = t^2$ is a finite variation is equivalent to showing the quadratic variation of $[t^2]_t = 0$, let $\pi = \{0 = t_0 < t_1 < \cdots < t_n = t\}$ be a partition with mesh $|\pi| = \max_i(t_{i+1} - t_i)$. Since

   $$(t_{i+1} - t_i)^2 \le |\pi|\,(t_{i+1} - t_i),$$

   we obtain

   $$\sum_i (t_{i+1} - t_i)^2 \le |\pi| \sum_i (t_{i+1} - t_i) = t\,|\pi|.$$

   As $|\pi| \to 0$, the right-hand side tends to 0, hence

   $$[t^2]_t = 0.$$

   Therefore, the only contribution to the quadratic variation comes from the local martingale part W_t. Since

   $$[W]_t = t,$$

   we conclude that

   $$[X]_t = t.$$

2. If X is a semimartingale and H bounded predictable, prove $\int_0^t H_s dX_s$ is a semimartingale.
Solution:

Since X is a semimartingale, write $X = M + A$ where M is a local martingale and A a finite-variation process.

Then:
$$\int_0^t H_s \, dX_s = \int_0^t H_s \, dM_s + \int_0^t H_s \, dA_s.$$

The first term is a local martingale because H is bounded and predictable. The second term has finite variation because A has finite variation and H is bounded. Thus, the integral is a semimartingale.

3. Suppose $X = M + A = M' + A'$ are two decompositions. Prove $M = M'$ using the fact that $M - M'$ is both local martingale and finite variation.
Solution:

From the two decompositions,
$$M - M' = A' - A.$$

Let $N = M - M'$. Then N is a local martingale and also a finite-variation process.

A continuous finite-variation local martingale must be constant. Taking $M_0 = M_0' = 0$, we get $N_0 = 0$, hence $N_t = 0$ for all t.

Therefore $M = M'$ and $A = A'$.

4. Verify that the compensated Poisson process $M_t = N_t - \lambda t$ is a martingale and N_t is a semimartingale.(see Section 4.5, Example 3)

5. Show that the product $X_t Y_t$ of two semimartingales satisfies the integration-by-parts formula.
Solution:

Let $X = M^X + A^X$ and $Y = M^Y + A^Y$ be their decompositions.

The formula states:
$$d(X_t Y_t) = X_t \, dY_t + Y_t \, dX_t + d[X, Y]_t.$$

Expanding:
$$d(XY) = d(M^X + A^X)(M^Y + A^Y).$$

Compute term by term:
$$d(M^X M^Y) = M^X dM^Y + M^Y dM^X + d[M^X, M^Y],$$
$$d(M^X A^Y) = M^X dA^Y + A^Y dM^X,$$
$$d(A^X M^Y) = A^X dM^Y + M^Y dA^X,$$
$$d(A^X A^Y) = A^X dA^Y + A^Y dA^X.$$

Adding all terms:
$$d(XY) = X \, dY + Y \, dX + d[M^X, M^Y].$$

Since $[X, Y] = [M^X, M^Y]$, we obtain:
$$d(XY) = X \, dY + Y \, dX + d[X, Y].$$

This is the integration-by-parts formula.

4.11 Summary and connections

Class of process	**Defining property / Decomposition**
Martingale	$\mathbb{E}[M_t\|\mathcal{F}_s] = M_s$, $A \equiv 0$
Local martingale	Exists localizing sequence (τ_n) with M^{τ_n} martingales
Semimartingale	$X = M + A$ with M local martingale, A finite variation
Finite-variation process	$A_t = \int_0^t a_s ds + \sum_j \Delta A_{t_j}$ with bounded variation
Continuous semimartingale	X continuous, A continuous finite var, M continuous loc mart.

Semimartingales unify the theory: they are stable under smooth transformations, addition, stopping, stochastic integration, and appear as the driving processes of general stochastic differential equations. They form the mathematical backbone of stochastic calculus.